Talha Khan

(734) 740 1486 | EE.TalhaKhan@gmail.com | Portfolio: TalhaAKhan2006.github.io | Linkedin.com/in/TalhaKhan-ee

SUMMARY

Electrical Engineering student at the University of Michigan–Dearborn with experience leading multidisciplinary engineering teams developing autonomous vehicle systems. Skilled in embedded systems, electrical integration, and wire harness design while managing a $30,000 engineering project.

EDUCATION

University of Michigan–Dearborn — **Dearborn, MI**
BSE in Electrical Engineering | *GPA:* ***3.87*** — *May 2027 (Expected)*

Wayne County Community College — **Belleville, MI**
Associate of Science | *GPA:* ***3.95*** — *May 2024*

TECHNICAL SKILLS

Programming: Python, C, JavaScript, HTML/CSS
Engineering: Embedded Systems, Wire Harness Design, Electrical Integration, Circuit Design, Power Distribution, Autonomous Systems, PCB Design
Software: Fusion 360, Altium Designer, LTspice, AutoCAD, Linux, Git
Hardware: Oscilloscopes, Function Generators, Multimeters, Soldering, Arduino, STM32, HCS12

EXPERIENCE

Chief Engineer & Treasurer
Intelligent Systems Club — Project Phoenix — *Aug 2025 – Present*

- Lead the electrical, mechanical, and software integration of an autonomous go-kart that earned **3rd Place** at the Autonomous Karting Series (AKS) National Grand Prix.
- Manage a $30,000 project budget, overseeing procurement, planning, and resource allocation while keeping the project under budget and on schedule.
- Direct multidisciplinary engineering efforts across a 20-member student team through design reviews,integration, and project planning.

Teaching Assistant (Grader)
University of Michigan–Dearborn — ECE 270, ECE 273, ECE 3731 — *Jan 2025 – Present*

- Independently managed feedback to students by addressing questions, clarifying course material, and providing constructive guidance to improve understanding.
- Review and grade assignments, quizzes, and examinations for ECE courses while providing technical feedback on circuits, embedded systems, and programming.

Mentor
FIRST Robotics Team 6190 — *Sept 2021 – May 2026*

- Mentored a 20+ member FIRST Robotics team in CAD, electrical wiring, fabrication, and system integration.
- Taught industrial safety standards, electrical wiring, and pneumatics systems to high school team members with zero prior engineering experience

PROJECTS

Project Phoenix
Autonomous Go-Kart — *2024 – Present*

- Designed and integrated the vehicle's electrical wire harness.
- Developed electrical routing, termination, and power distribution.
- Integrated embedded sensors and motor controllers during system validation.

Autonomous Mobile Robot
Fall 2024

- Designed a line follower AMR with obstacle avoidance capabilities.
- Goal was to achieve speed and stability while carrying a load.
- Arduino Framework

Adaptive Light System
ECE 210 — *Fall 2025*

- Designed an automatic lighting system using a CdS photoresistor and op-amp comparator.
- Reducing estimated energy consumption by 30–35%.

AWARDS & ACTIVITIES

Dean's List | Emerging Scholar | Mentor Collective Certification (Diversity, Equity & Inclusion) | Tau Beta Pi Engineering Honor Society | IEEE | Intelligent Systems Club | Phi Theta Kappa | Muslim Student Association